

Name of experiment : Design and implement a Binary-to-gray code converter.

Objective: To study of binary to gray code conversion.

Theory: The gray code has been developed as a way to represent a sequence of numbers. The unique aspect of the Gray code is that only one bit ever changes between two successive numbers in the sequence.

Binary-to-gray Conversion:

In this conversion, the input straight binary number can easily be converted to its Gray code equivalent.

- ① Record the most significant bit as it is.
- ② Ex-OR this bit to the next position bit, record the resultant bit.
- ③. Record successive x-OR bits until completed.

4. Convert 0011 binary to Gray

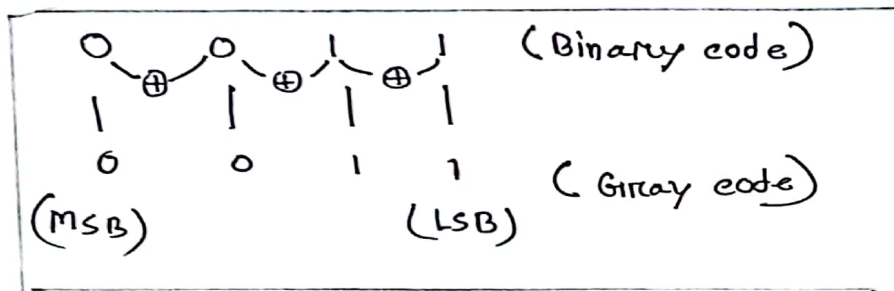


fig: Binary to gray code conversion

Logical circuit Diagram for Binary to Gray code conversion.

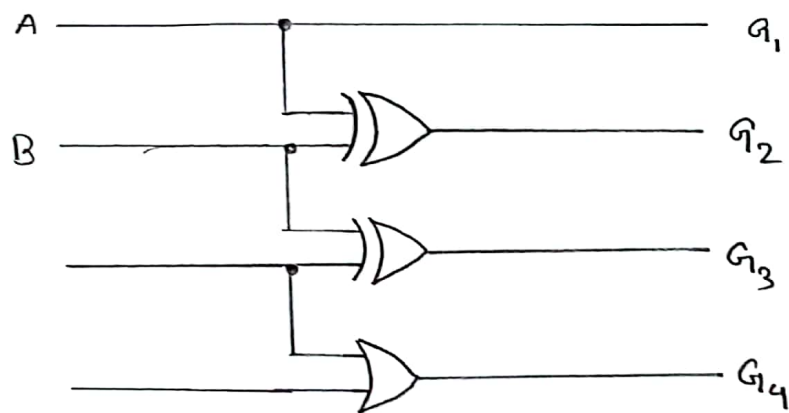


fig: Circuit Diagram

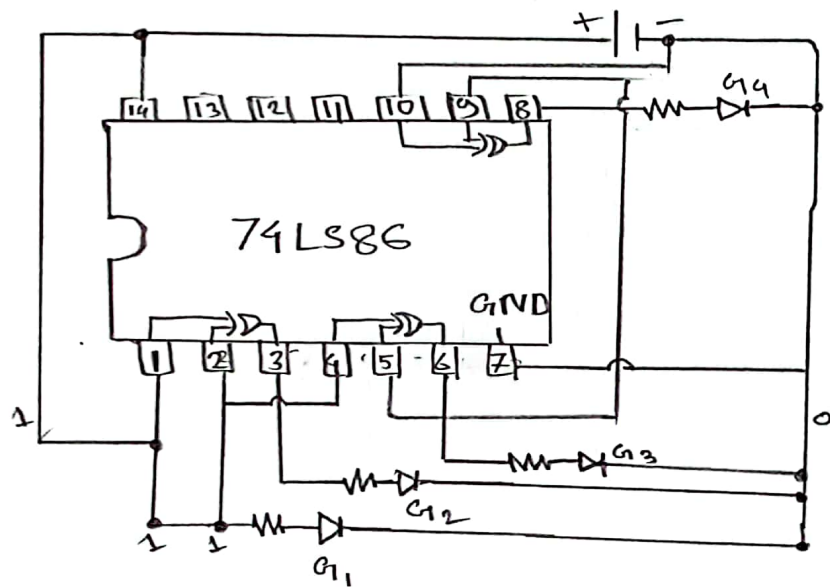
Truth table:

Decimal number	4-bit Binary code				4-bit Gray code			
	A	B	C	D	G ₁	G ₂	G ₃	G ₄
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	0
5	0	1	0	1	0	1	1	1
6	0	1	1	0	0	1	0	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	0	0
9	1	0	0	1	1	1	0	1
10	1	0	1	0	1	1	1	1
11	1	0	1	1	1	1	1	0
12	1	1	0	0	1	0	1	0
13	1	1	0	1	1	0	1	1
14	1	1	1	0	1	0	0	1
15	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0

Apparatus:

- ① Breadboard
- ② X-OR gate (74LS86)
- ③ LED
- ④ Resistor
- ⑤ Power supply
- ⑥ wire.

Procedure: At first we have to identify the pin numbers of the IC and then set it on the breadboard. Connected all the V_{cc} with the wire positive power supply and all the ground with negative power supply. Finally connect all the input and output wires with resistor and LED.



Result

A	B	C	D	G ₁	LED	G ₂	LED	G ₃	LED	G ₄	LED
0	0	0	1	0	off	0	off	0	off	1	on
0	0	1	0	0	off	0	off	1	on	1	on
0	0	1	1	0	off	0	off	1	on	0	off
0	1	0	0	0	off	1	on	1	on	0	off
0	1	0	1	0	off	1	on	1	on	1	on
0	1	1	0	0	off	1	on	0	off	1	on
0	1	1	1	0	off	1	on	0	off	0	off
1	0	0	0	1	on	1	on	0	off	0	off
1	0	0	1	1	on	1	on	0	off	1	on
1	0	1	0	1	on	1	on	1	on	1	on
1	0	1	1	1	on	1	on	1	on	0	off
1	1	0	0	1	on	0	off	1	on	0	off
1	1	0	1	1	on	0	off	1	on	1	on
1	1	1	0	1	on	0	off	0	off	1	on
1	1	1	1	1	on	0	off	0	off	0	off
0	0	0	0	0	off	0	off	0	off	0	off

Discussion: In this experiment we have checked all the gates of ICs and found the active gates. we found out the position here the IC got proper supply and use these for this experiment.